

Modeling with Systems and Matrices

Modeling with two- and three-variable systems, writing them as matrix equations, and finding where a line meets a curve

- Define your variables in words before you write any equation, and say what unit each one carries.
 - The method is yours to choose: substitution, elimination, or a matrix equation. Pick the one the numbers make easiest, and say which you used.
 - A system with a squared term can have two solutions. Find both, then decide which ones the situation allows.
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1. A theatre sells adult tickets at x dollars each and child tickets at y dollars each. One family buys 3 adult and 2 child tickets for 61 dollars. Another buys 1 adult and 4 child tickets for 47 dollars. Find x and y .

Answer: _____

2. A farm stand ships crates and boxes. Four crates and three boxes together weigh 47 kilograms. Two crates weigh 9 kilograms less than five boxes. Let x be the mass of one crate and y the mass of one box, both in kilograms. Find x and y .

Answer: _____

3. A roaster blends two coffees to make 20 kilograms of a house mix. The first coffee costs 12 dollars per kilogram and the second costs 18 dollars per kilogram; the mix must cost 288 dollars in all. Let x and y be the kilograms of each coffee. Write the situation as a matrix equation, then find x and y .

Answer: _____

4. A water jet from a fountain follows the path $y = -x^2 + 8x + 9$, and a straight support wire follows $y = 3x + 15$, with x the horizontal distance and y the height, both in metres. Find every point where the jet meets the wire.

Answer: _____

5. A workshop made 16 items last week: chairs, tables and stools. Cutting takes 2 hours per chair, 3 hours per table and 1 hour per stool, and used 34 hours in all. Finishing takes 3 hours per chair, 1 hour per table and 2 hours per stool, and used 29 hours in all. Let x , y and z be the number of chairs, tables and stools. Find x , y and z .

Answer: _____

6. A juice bar mixes orange juice at 3 dollars per litre with mango juice at 5 dollars per litre. The batch is 24 litres and costs 88 dollars. Let x and y be the litres of orange and mango juice.

(a) Write the situation as a matrix equation and find x and y .

Answer: _____

(b) Explain what each *row* of your coefficient matrix records about the batch.

Answer: _____

7. A circular pond has edge $x^2 + y^2 = 25$, where x and y are distances in metres from the centre of the pond. A straight gravel path runs along $y = x + 1$. Find both points where the path crosses the pond edge.

Answer: _____

8. A workshop rents a floor sander two ways. Plan A charges 40 dollars plus 6 dollars per hour; Plan B charges 25 dollars plus 9 dollars per hour. Let x be the hours rented and y the total cost in dollars.

(a) Find the number of hours at which the two plans cost the same, and that shared cost.

Answer: _____

(b) Explain which plan is cheaper for a 10-hour rental, and why.

Answer: _____

9. A cafe sold 20 drinks in one hour: small, medium and large. Smalls cost 3 dollars, mediums 4 dollars and larges 5 dollars, and the hour brought in 79 dollars. The cafe sold 3 more mediums than smalls. Let x , y and z be the numbers of small, medium and large drinks. Find x , y and z .

Answer: _____

10. A rectangular garden has a perimeter of 34 metres and an area of 60 square metres. Let x and y be its two side lengths in metres.

(a) Find every pair (x, y) that satisfies both conditions.

Answer: _____

(b) Explain why the two pairs you found describe the same garden.

Answer: _____

11. A feed mill blends three grains into a 100 kilogram mix. Per kilogram, the grains supply 2, 1 and 3 units of protein, and the mix must supply 220 units. Per kilogram they supply 3, 2 and 1 units of fibre, and the mix must supply 170 units. Let x , y and z be the kilograms of each grain. Write the blend as a matrix equation and find x , y and z .

Answer: _____

12. A rectangular field is fenced on three sides, with a river forming the fourth. The fencing available is 120 metres and the enclosed area must be 1600 square metres. Let x be the side parallel to the river and y each of the two sides running back from it, both in metres.

(a) Find every pair (x, y) that uses all the fencing and encloses the required area.

Answer: _____

(b) Which pair gives the wider river frontage? Explain, and say whether both pairs are genuinely usable designs.

Answer: _____